

COURSE DESCRIPTION			
Program	Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Civil Engineering (Construction Technology)		
Course Title	Basic Surveying		
Course Code	2012	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

Basic Surveying introduces the fundamental principles and methods used for measurement and mapping in civil engineering works. The course covers chain and tape surveying, basic compass and plane table concepts, levelling, contouring basics, and introductory theodolite surveying. It provides a foundation for advanced surveying and practical field applications.

Course Objectives

1	To develop competency in chain and tape surveying techniques and simple area calculations
2	To Compute reduced levels using standard levelling methods
3	Solve numerical problems related to levelling applications and contour basics
4	To familiarize students with contouring basics and introductory theodolite surveying.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in mathematics	1002	Fundamentals of Engineering Mathematics	I
Basic principles and purpose of surveying	1011	Fundamentals of building construction	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	To develop competency in chain and tape surveying techniques and simple area calculations.	Understand
CO2	Compute reduced levels using standard levelling methods	Apply
CO3	Solve numerical problems related to levelling applications and contour basics	Apply
CO4	Explain basic theodolite concepts and methods of horizontal angle measurement	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	1	-	-	-	-	-	-
CO2	3	3	2	2	1	-	-	-	-	-	-
CO3	3	3	2	1	1	-	-	-	-	-	-
CO4	2	1	2	2	-	-	-	-	-	-	-
Total	11	9	8	6	3	-	-	-	-	-	-
Correlation Level	2.75	2.25	2	1.5	1	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Chain & Tape Surveying, Compass and Plane Table - Introduction	Chain and tape surveying: Instruments and accessories (chain and tape), types of chain and tapes (names only), selection of stations, base line, tie line, check line, ranging (brief), chaining and taping, taking offsets, cross staff survey and simple area calculation, conventional signs. Compass survey: Purpose, types of compass (names only), meridians (true and magnetic), bearings of a line, fore bearing and back bearing, whole circle bearing, meaning and calculation of included angle from bearings. Local attraction (brief) Plane table survey (Introduction only): Meaning and purpose, situations where plane table survey is used, advantages and disadvantages of plane table survey.	16	0
Module II	Levelling - Principles, Instruments and Reduction of Levels	Purpose of levelling. Definitions - level surface, datum, reduced level, benchmark. Types of bench mark (brief). Levelling instruments - dumpy, tilting and automatic level (Brief). Levelling staff - types. Temporary adjustments. Level book - format and field notes. Reduction of levels by height of collimation method and rise and fall method, comparison of methods and simple numerical problems.	16	0
Module III	Levelling Applications and Contour Survey - Basics	Types of levelling - fly levelling, profile levelling, check levelling, reciprocal levelling Numerical problems related to types of levelling - computation of reduced levels, checks and simple gradient calculations. Longitudinal section and cross section - basic concept and plotting. Contour survey - definition of contour, characteristics of contours, contour interval, uses of contour maps.	16	0
Module IV	Introduction to Theodolite Surveying	Introduction to theodolite. Types of theodolites (outline). Main parts of a transit theodolite and their functions. Temporary adjustments of a theodolite. Measurement of horizontal angles - general procedure, method of repetition, and method of reiteration. Applications of theodolite in civil engineering works.	12	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Chain & tape instruments	Identify chain and tape instruments	L	CO1	Remember	2
Survey lines	Explain base line, tie line and check line	L	CO1	Understand	2
Ranging	Apply direct and indirect ranging	L	CO1	Understand	2
Chaining & offsets	Apply chaining and offset methods	L	CO1	Understand	1
Conventional signs	Identify conventional signs used in surveying	L	CO1	Remember	1
Area calculation	Compute simple areas from field data	L	CO1	Apply	2
Compass survey - Introduction	Explain Meridians, bearings, Fore bearing, back bearing, WCB, QB	L	CO1	Understand	2
Included angle, local attraction	Calculation of included angles from bearings and concept of local attraction	L	CO1	Apply	2
Plane Table Survey (Introductory)	Identify applications, advantages and limitations of plane table survey	L	CO1	Understand	2
Module II					
Basic terms	Explain level surface, datum and Reduced level	L	CO2	Understand	2
Bench marks	Identify types of bench marks	L	CO2	Remember	1
Levelling instruments	Identify types of levelling instruments	L	CO2	Remember	2
Levelling staff	Explain types of levelling staff	L	CO2	Remember	2
Temporary adjustments	Explain levelling adjustments	L	CO2	Understand	2
Level book	Explain recording of field data	L	CO2	Understand	2

Height of collimation method	Compute RLs using HI method	L	CO2	Apply	2
Rise & fall method and comparison	Compute RLs and compare methods	L	CO2	Apply	3
Module III					
Fly levelling and differential levelling	Explain fly levelling and differential levelling procedure	L	CO3	Understand	2
Profile levelling and check levelling	Explain profile levelling and check levelling	L	CO3	Understand	2
Reciprocal levelling	Explain reciprocal levelling	L	CO3	Understand	2
Levelling numerical	Solve levelling application problems	L	CO3	Apply	2
Gradient calculation	Compute gradients	L	CO3	Apply	2
Longitudinal sectioning and cross sectioning	Explain L-section plotting and C-section plotting	L	CO3	Apply	2
Contours – definition & characteristics	Explain contour concepts and properties	L	CO3	Understand	2
Uses of contour maps	Interpret contour maps	L	CO3	Understand	2
Module IV					
Introduction	Explain basic theodolite concepts	L	CO4	Understand	2
Types	Identify types of theodolites	L	CO4	Remember	1
Main parts & functions of theodolite	Identify parts of a transit theodolite	L	CO4	Remember	2
Temporary adjustments	Explain theodolite setup procedure	L	CO4	Understand	2
Permanent adjustment	Fundamental lines and their relationship concepts	L	CO4	Understand	2
Horizontal angle measurement	Methods of measuring horizontal angles, by general procedure, method of repetition, and method of reiteration.	L	CO4	Understand	2
Applications in Civil engineering	Explain uses of theodolite	L	CO4	Understand	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Study chain and tape instruments with labeled diagrams and note their uses.	6
2	Practice numerical problems on chain surveying and simple area calculation.	6
3	Prepare notes on bearings, fore bearing, back bearing and included angles.	6
4	Watch videos/read material on plane table survey methods	6
Module II		
5	Prepare glossary of levelling terms and bench marks with neat diagrams.	6
6	Study levelling instruments and temporary adjustments..	6
7	Solve numerical problems using Height of Collimation method.	6
8	Solve numerical problems using Rise and Fall method and compare results.	6
Module III		
9	Study fly, profile and reciprocal levelling procedures.	6
10	Practice gradient calculations and levelling application problems.	6
11	Prepare longitudinal and cross section sketches.	6
12	Study contour characteristics and interpret contour maps.	6
Module IV		
13	Study parts of theodolite and prepare labelled diagram.	6
14	Prepare step-wise notes on horizontal angle measurement methods.	6
15	Study applications of theodolite in civil engineering works.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Surveying and Levelling	T. P. Kanetkar & S. V. Kulkarni	Pune Vidyarthi Griha Prakashan
2	Surveying – Volume I & II	B. C. Punmia	Laxmi Publications

REFERENCE			
Sl. No.	Title	Author	Publication
1	A Textbook of Surveying and Levelling	R. Agor	Khanna publishers
2	Surveying and levelling	R. Subramanian	Oxford Higher Education

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in	Levelling, Contours, Theodolite
2	https://www.civilenggforall.com	Chain survey, Compass survey
3	https://www.engineeringcivil.com	All

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60